



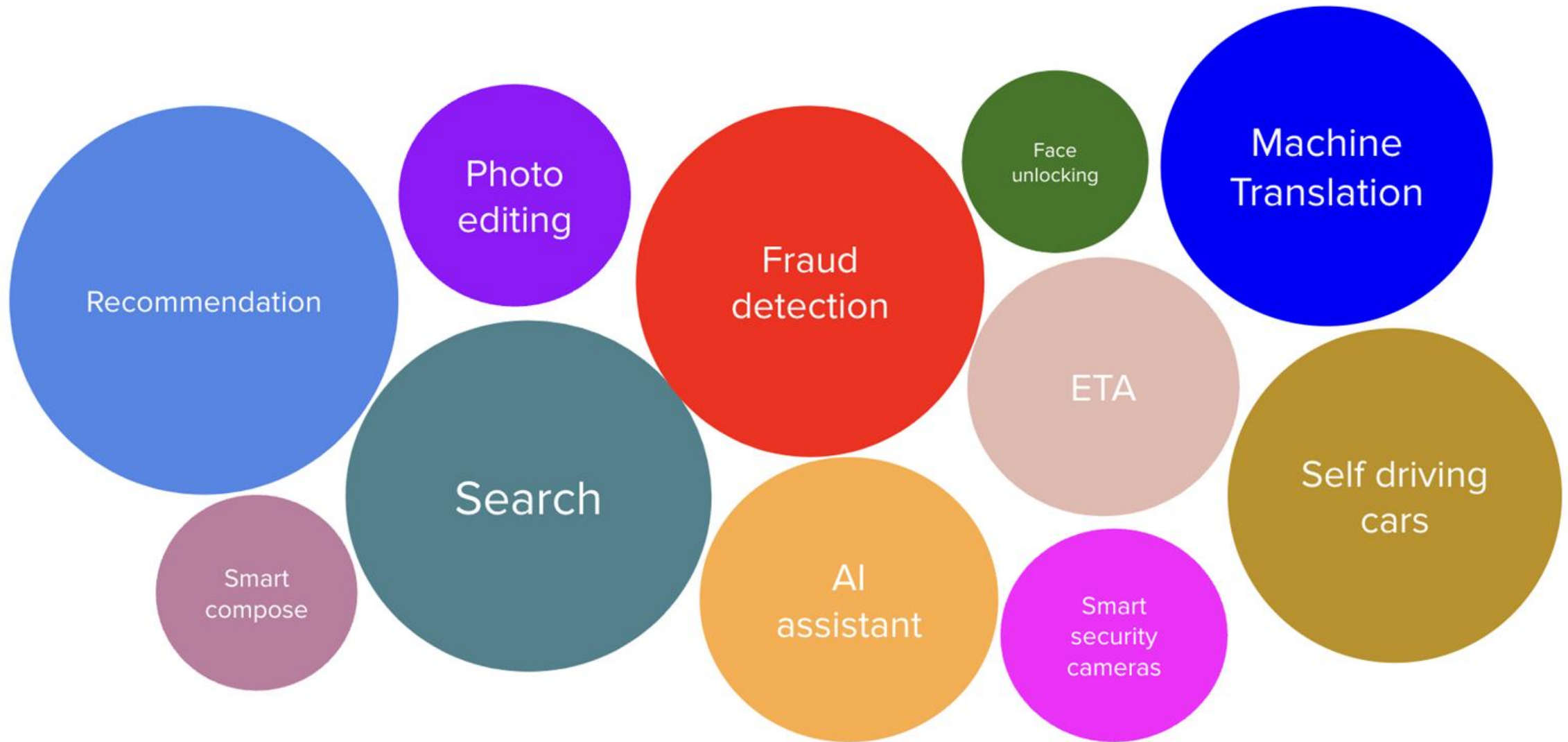
**SHARIF**  
UNIVERSITY OF  
TECHNOLOGY

# Foundations of Data Science Introduction (Part 2)

**Sharif University of Technology**  
**Department of Electrical Engineering**

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Fall 2026

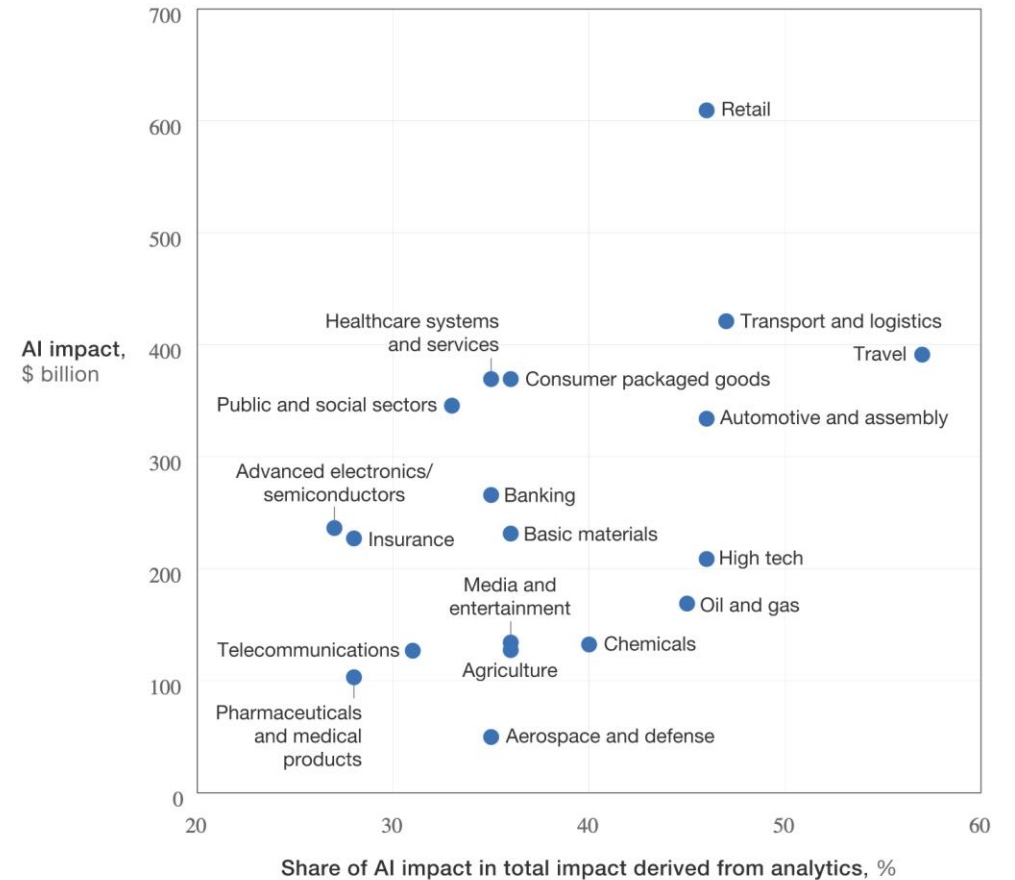
# ML is in Almost Every Aspect of Our Lives



# Market of Machine Learning and Artificial Intelligence

- AI value creation by 2030 is estimated to be **13 trillion USD!**
- Most of it will be outside the consumer internet industry
- **We need more people from non-CS background in AI!**

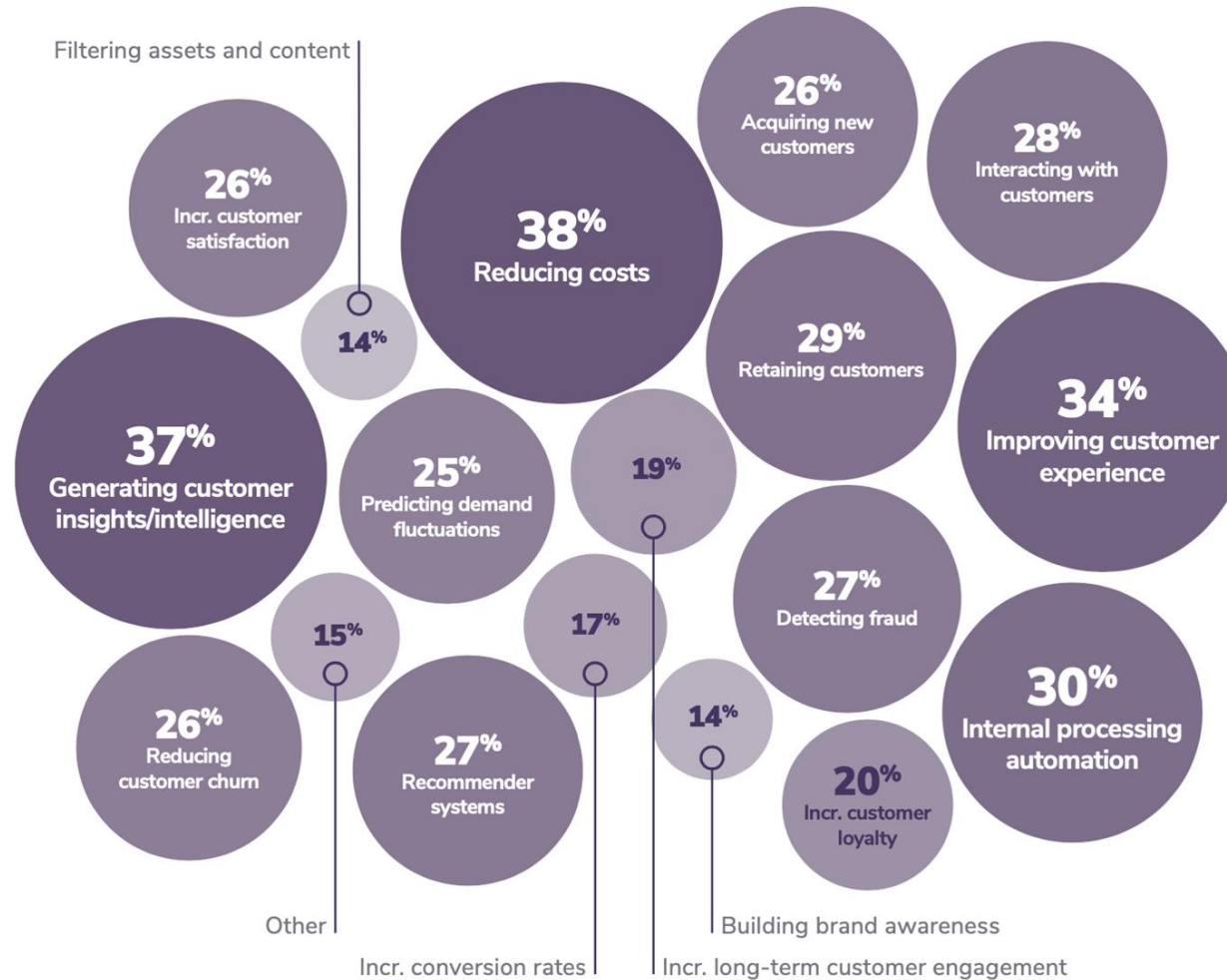
Artificial intelligence (AI) has the potential to create value across sectors.



McKinsey&Company | Source: McKinsey Global Institute analysis

# Enterprise Use Cases of ML

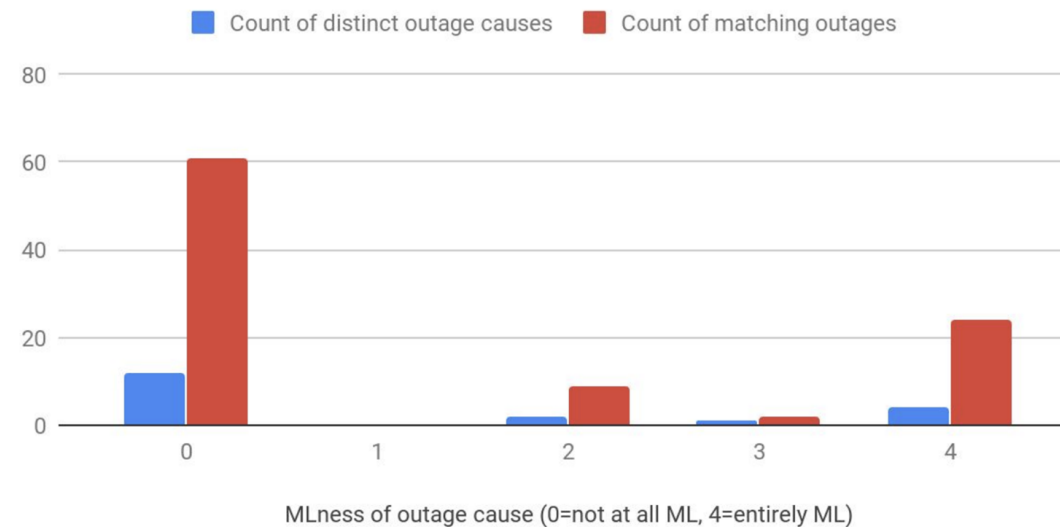
Machine learning use case frequency



# Why ML Systems Design?

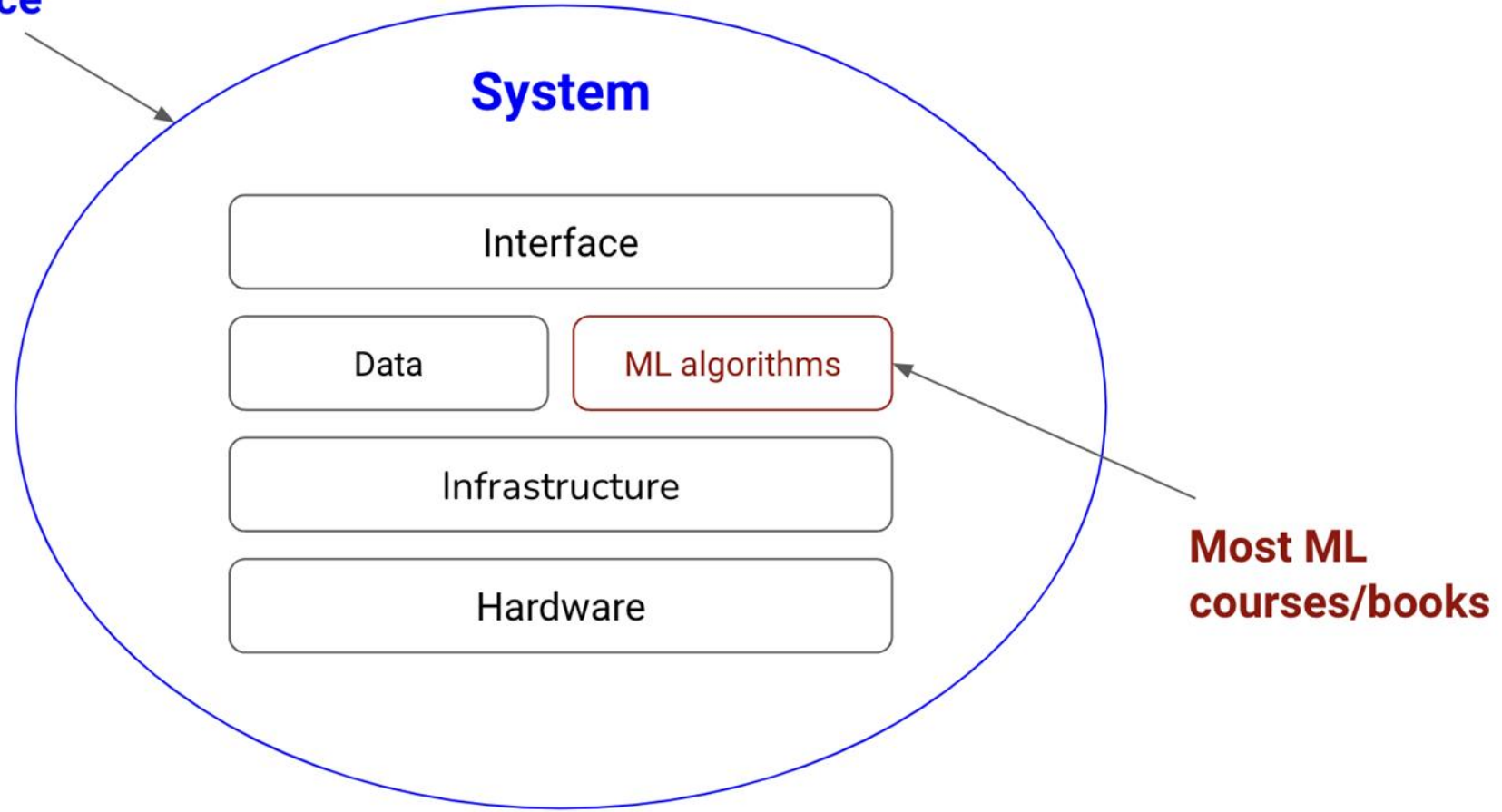
- ML algorithms is the less problematic part.
- The hard part is to how to make algorithms work with other parts to solve real-world problems.
- 90/96 percent of failures caused by non-ML components

Count of distinct outage causes and Count of matching outages



# Our Course Vs. The Usual Suspects

Foundations of  
Data Science



# What is Machine Learning Systems' Design?

- The process of defining the **interface, algorithms, data, infrastructure**, and **hardware** for a machine learning system to satisfy **specified requirements**.
- Specified requirements means systems are **reliable, maintainable, scalable, and adaptable** to our needs.

# The Questions that Will Arise and Will be Answered...

- You've trained a model, now what?
- What are different components of an ML system?
- How to do data engineering?
- How to engineer features?
- How to evaluate your models, both offline and online?
- What's the difference between online prediction and batch prediction?
- How to serve a model on the cloud? On the edge?
- How to continually monitor and deploy changes to ML systems?
- ...

# ML Research Vs. ML Production

|            | <b>Research</b>    | <b>Production</b>                                |
|------------|--------------------|--------------------------------------------------|
| Objectives | Model performance* | Different stakeholders have different objectives |

# Stakeholder Objectives

## Sales Team

Wants to sell more ads

## ML team

Wants the highest accuracy (The same as when you look at ML from theoretical side)

## Production Team

Fastest inference to enable scaling and minimize costs

## Manager

Wants to maximize profit and it might mean letting the ML team go :)

# Computational Priority

|                        | Research                       | Production                                       |
|------------------------|--------------------------------|--------------------------------------------------|
| Objectives             | Model performance              | Different stakeholders have different objectives |
| Computational priority | Fast training, high throughput | Fast <b>inference</b> , low latency              |

- 100ms delay can hurt conversion rates by 7% ([Akamai study](#) '17)
- 30% increase in latency costs 0.5% conversion rate ([Booking.com](#) '19)
- 53% phone users will leave a page that takes >3s to load ([Google](#) '16)

# Latency and Throughput. What are They?

Let's explain latency and throughput in simple terms. Consider a bird moving leaves to its nest.

- Latency: time to move a leaf
- Throughput: how many leaves in 1 sec
- Real-time: low latency = high throughput  
(We need this for applications in production e.g. Snapp, Divar)
- Batched: high latency, high throughput  
(We can use this for lots of data that don't change that much)

# Computational Priority

|                        | <b>Research</b>                | <b>Production</b>                                |
|------------------------|--------------------------------|--------------------------------------------------|
| Objectives             | Model performance              | Different stakeholders have different objectives |
| Computational priority | Fast training, high throughput | Fast inference, low latency                      |
| Data                   | Static                         | Constantly shifting                              |

# Data in Research Versus Data in Production

| Research                                                                                                  | Production                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Clean</li><li>• Static</li><li>• Mostly historical data</li></ul> | <ul style="list-style-type: none"><li>• Messy</li><li>• Constantly shifting</li><li>• Historical + streaming data</li><li>• Biased, and you don't know how biased</li><li>• Privacy + regulatory concerns</li></ul> |

## THE COGNITIVE CODER

By **Armand Ruiz**, Contributor, InfoWorld | SEP 26, 2017 7:22 AM PDT

## The 80/20 data science dilemma

Most data scientists spend only 20 percent of their time on actual data analysis and 80 percent of their time finding, cleaning, and reorganizing huge amounts of data, which is an inefficient data strategy

# Fairness in Research and Production

|                        | <b>Research</b>                | <b>Production</b>                                |
|------------------------|--------------------------------|--------------------------------------------------|
| Objectives             | Model performance              | Different stakeholders have different objectives |
| Computational priority | Fast training, high throughput | Fast inference, low latency                      |
| Data                   | Static                         | Constantly shifting                              |
| Fairness               | Good to have (sadly)           | Important                                        |

# Now What is Fairness?



## Google Shows Men Ads for Better Jobs

by Krista Bradford | Last updated Dec 1, 2019

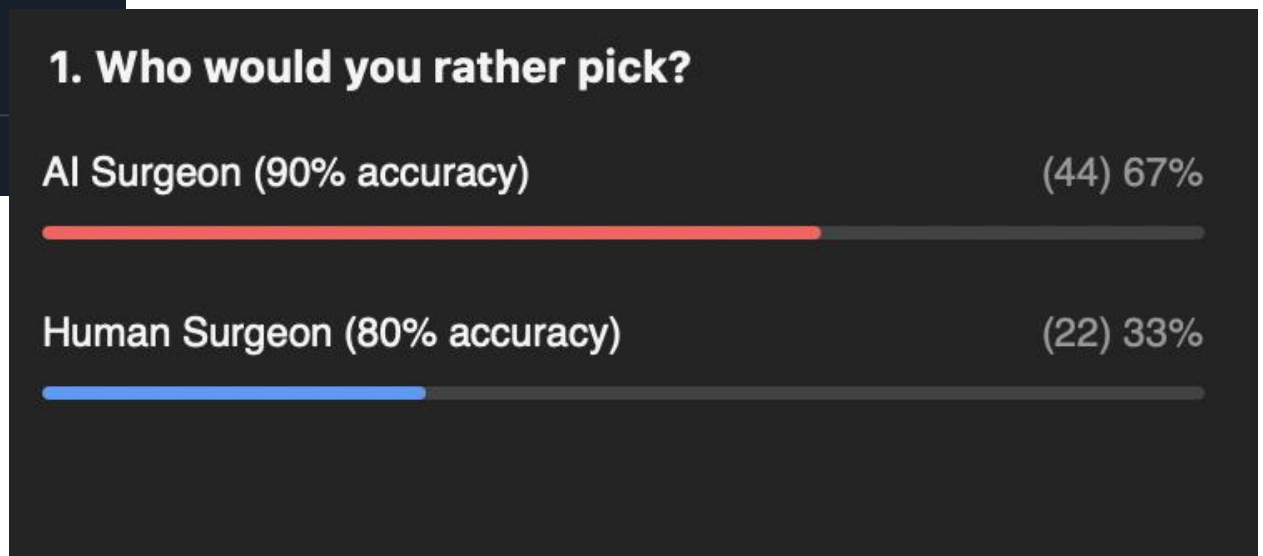
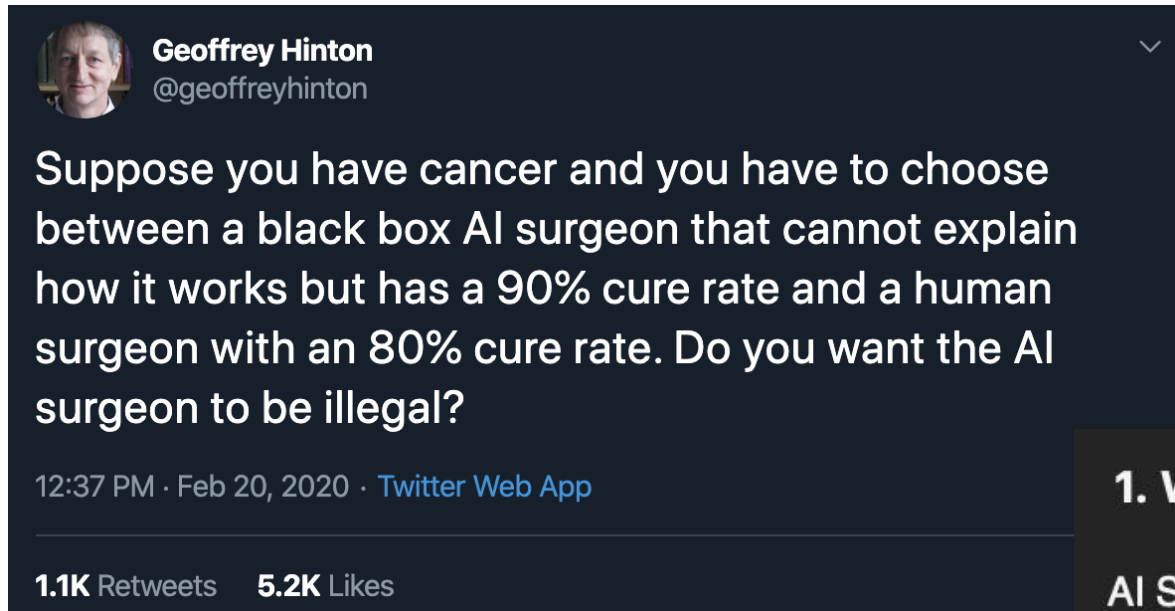


The Berkeley study found that both face-to-face and online lenders rejected a total of 1.3 million creditworthy black and Latino applicants between 2008 and 2015. Researchers said they believe the applicants "would have been accepted had the applicant not been in these minority groups." That's because when they used the income and credit scores of the rejected applications but deleted the race identifiers, the mortgage application was accepted.

# Interpretability of ML Projects

|                        | <b>Research</b>                | <b>Production</b>                                |
|------------------------|--------------------------------|--------------------------------------------------|
| Objectives             | Model performance              | Different stakeholders have different objectives |
| Computational priority | Fast training, high throughput | Fast inference, low latency                      |
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| Fairness               | Good to have (sadly)           | Important                                        |
| Interpretability*      | Good to have                   | Important                                        |

# What is Interpretability?



# This was Research vs. Production.

## Now Let's Talk about Production.

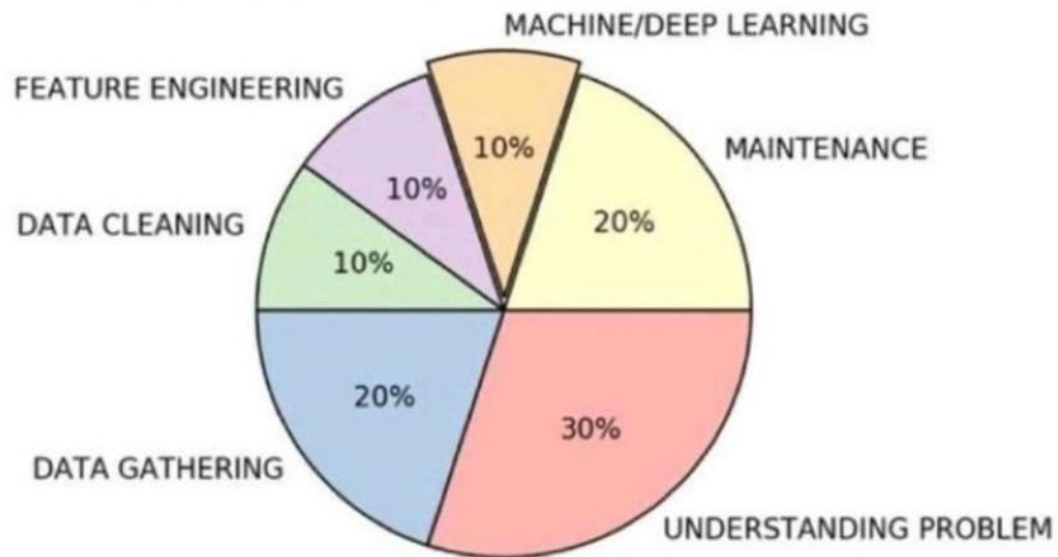
- What you think it is
  - Collect data
  - Train model
  - Deploy model



- Reality
  - Choose a metric to optimize
  - Collect data
  - Train model
  - Realize many labels are wrong > Relabel data
  - Train model
  - Model performs poorly > Maybe collect more data
  - Train model
  - Deploy model
  - Biases against older groups
  - Get more data, train model, do more testing
  - Deploy model
  - Performs well for a while, Do the process again

# Data Scientist Expectation Vs. Reality

**DATA SCIENTIST JOB - REALITY**



**DATA SCIENTIST JOB - EXPECTATION**



# Real World Project Data Science Objectives

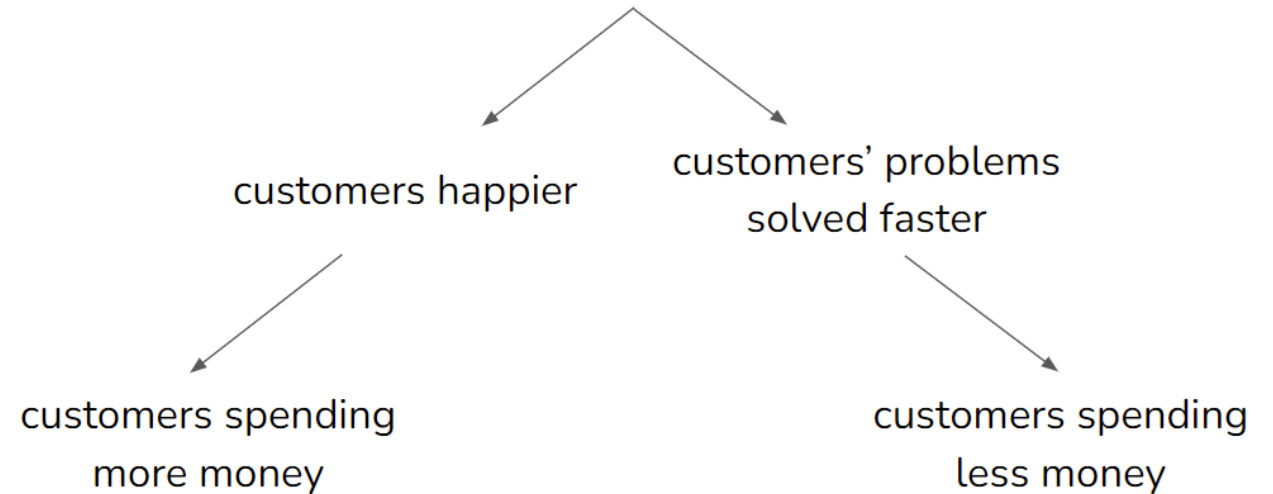
- **Two main types of objectives**

- ML objectives
  - Performance
  - Latency
- Business objectives
  - Cost
  - ROI
  - Regulation & compliance

- **Most important business objective:**

- How can this ML project increase profits directly or indirectly?
  - Directly: increasing sales (ads, conversion rates), cutting costs
  - Indirectly: increasing customer satisfaction, increasing time spent on a website

ML model gives customers more personalized solutions



# Business Considerations in a Data Science Project

- Baselines
  - Existing solutions, simple solutions, human experts, competitors solutions, etc.
- Usefulness threshold
  - Self-driving needs human-level performance. Predictive texting doesn't.
- False negatives vs. false positives
  - Covid screening: no false negative (patients with covid shouldn't be classified as no covid)
  - Fingerprint unlocking: no false positive (unauthorized people shouldn't be given access)
- Interpretability
  - Does it need to be interpretable? If yes, to whom?
- Confidence measurement (how confident it is about a prediction)
  - Does it need confidence measurement? Is there a confidence threshold?

# Constraints

- Time
  - Rule of thumb: 20% time to get initial working system, 80% on iterative development
- Budget
  - Data, resources, talent
- Time/budget trade offs
  - Use more (powerful) machines
  - Hire more people to label data faster
  - Run more experiments in parallel
  - Buy existing solutions

# Constraints

- Privacy
  - Annotation
    - Can data be shipped outside organizations for annotation?
  - Storage
    - What kind of data are you allowed to store? How long can you store it?
  - Third-party solutions
    - Can you share your data with a 3rd party (e.g. managed service)?
  - Regulations
    - What regulations do you have to conform to?
- Technical
  - Competitors and Legacy Systems

# Machine Learning Phases

## The 4 phases of ML adoption

1. Before ML
  1. “If you think that machine learning will give you a 100% boost, then a heuristic will get you 50% of the way there.”, Martin Zinkevich, Google
2. Simplest ML Model
  1. Start with a simple model that allows visibility into its working to:
    1. validate hypothesis and pipelines
3. Optimizing simple models
  1. Different objective functions
  2. Feature engineering
  3. More data
4. Complex ML models

# Summary of What We Learned In This Session.

- ML in different aspects of our lives
- Market of Machine Learning
- Use cases of Machine learning
- Difference between our course and other courses
- Machine learning systems' design
- ML research versus ML production

**-Objectives**

**-Computational priority**

**-Data**

**-**

**Fairness**

**- Interpretability**

- ML in production. Expectation vs. reality

Constraints and machine learning phases